Page 1 | Headline, population and comparisons

Trainer-held until requested. These source statements concern a fictional case.

E1: Which estimate is the headline?

Supplied evidence. The note selected the pilot point estimate: 1,014 AED annual saving. The separate pilot has 400 businesses in one district, lottery assigned 200/200, with costs at 12 months. The comparison is lottery groups, not the city before/after mean.

Not supplied. No national representativeness or implementation-transfer evidence is supplied.

A useful follow-up. Send the lottery and follow-up protocol, plus evidence about the businesses and delivery conditions expected nationally.

Decision purpose. Assess the credibility of the pilot comparison and whether expansion changes the population or delivery conditions.

E2: Do all methods confirm the same effect?

Supplied evidence. The city file has 10,000 businesses. DiD compares changes; RDD fits a local jump using 545 businesses within score 58 +/- 2; matching compares 4,794 participant businesses with selected comparison businesses. These city analyses reuse records and estimate different comparisons. They are not independent replications.

Not supplied. No common national effect or genuinely independent city replication is supplied.

A useful follow-up. Identify the target population and preferred analysis, its rationale and any genuine independent evidence.

Decision purpose. Stop counting methods as votes; align the comparison and assumptions with the decision.

Source: corrected synthetic file and shared case analysis. E1 uses the separate pilot; E2 uses city records. No national-effect estimate is supplied.

Page 2 | Minimum saving and city trends

Trainer-held until requested. Use the plot if live R is unavailable.

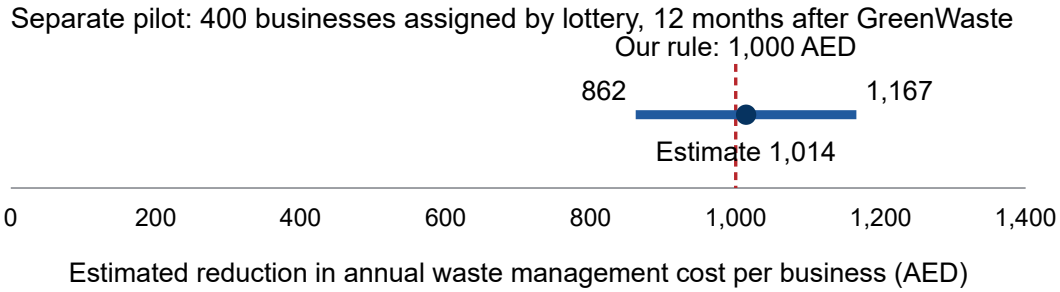
E3: Is the minimum saving settled?

Supplied evidence. On a positive annual-saving scale, the pilot estimate is 1,014 AED and its 95% interval is 862 to 1,167 AED. Pilot and matching intervals span the 1,000 AED rule; the DiD and RDD intervals lie below it. All four intervals lie above zero. Uncertainty about the minimum saving and evidence of a reduction are different questions.

Not supplied. No analysis resolves national threshold certainty. An interval alone cannot decide an acceptable risk level.

A useful follow-up. Show the estimate and interval alongside our rule of at least a 1,000 AED reduction; state the risk tolerance and evidence needed for a phased decision.

Decision purpose. Separate an estimated reduction from a settled minimum-saving claim.



E4: Can earlier city trends be checked?

Supplied evidence. The city file contains one before observation and one observation at 12 months per business. It cannot show earlier trends. The DiD causal interpretation requires the participant and comparison changes to stand in for what would have happened without support.

Not supplied. No earlier cost series or documented review of concurrent changes is supplied.

A useful follow-up. Send several earlier cost measures and an account of other changes affecting the two groups.

Decision purpose. Judge how plausible the counterfactual trend is; earlier trends cannot prove it.

E3: pilot OLS; DiD business-clustered; cutoff HC2; matching business-clustered with matches given. E4: one before measure in the supplied city file. Similar observed earlier trends, if obtained, would inform plausibility rather than prove the counterfactual.

Page 3 | Cutoff and matching checks

Trainer-held until requested.

E5: What else jumps at 58?

Supplied evidence. Businesses with a score at or below 58 took part. The local result uses a two-point window and an HC2 robust interval. A separate five-point diagnostic shows pre-programme managers about 5.3 years younger below the cut-off. This is a concrete comparability concern, not proof of which alternative cause explains the cost jump.

Not supplied. No score-manipulation audit or evidence resolving the age jump is supplied.

A useful follow-up. Send the score-setting audit, cut-off balance checks and sensitivity to reasonable windows/specifications.

Decision purpose. Assess continuity around the cut-off; a clean audit would not by itself establish every assumption.

E6: Are all matched businesses comparable?

Supplied evidence. Matching uses manager age, staff, area (in 100 square metres) and filtration before support. Every participant gets a nearest neighbour, without a distance limit. 585 distinct controls are used; one is reused 440 times. Residual gaps in mean manager age and in waste cost before GreenWaste are 3.0 years and 161 AED. The reported business-clustered interval treats the chosen matches as given.

Not supplied. No full closeness/support assessment or evidence ruling out unmeasured selection is supplied.

A useful follow-up. Send distance/overlap distributions, residual balance and sensitivity to features and unmeasured selection.

Decision purpose. Assess comparability and remaining selection; measured balance alone cannot rule out unmeasured bias.

E5: score-58 local window and pre-programme age discontinuity. E6: standardised manager age, staff, area and filtration; nearest neighbour with replacement, no caliper. Supplied gaps and reuse do not constitute a full support diagnostic.

Page 4 | Data quality and economic advice

Trainer-held until requested.

E7: Who is missing and how were outcomes measured?

Supplied evidence. These are complete generated records in a synthetic training file. No real recruitment, fieldwork, attrition, measurement-validation or representativeness documents are supplied. Completeness of this file does not demonstrate those qualities in an actual evaluation.

Not supplied. Real data-quality assurance cannot be inferred from the supplied file.

A useful follow-up. Send recruitment, follow-up by group, reasons for missing outcomes and measurement-validation records.

Decision purpose. Judge what the observed outcome comparison represents and whether measurement/follow-up could distort it.

E8: Does the ratio prove value for money?

Supplied evidence. The RCT-linked partial model gives a ratio of 2.44. Its annual benefit is the pilot saving estimate of 1,014 AED at 12 months. It assumes 1,800 AED delivery cost at year 0, equal annual savings in years 1-5 and a 5% rate. Delivery costs and later savings are not RCT measurements; this is not a complete national public-budget or social appraisal.

Not supplied. No measured full cost inventory or evidence establishing five-year persistence is supplied.

A useful follow-up. Send amounts, timing and perspective for all relevant costs, persistence evidence and sensitivity results.

Decision purpose. Judge a spending recommendation without turning assumed inputs into measured facts.

E7: synthetic-record completeness is not a real-world fieldwork audit. E8 revisits the RCT-linked 2.44 partial model from Day 3 Session 1. Its cost and later savings are assumptions; the saving rule and a benefit-cost ratio answer different questions.